THE AI TOOLBOX · CLASS 09

# Data Visualization (Data Storytelling)

Use charts to **argue**, not just to decorate.

Merit Point Academy · 60 min · press **S** notes · → next · 💬 to comment

RECAP · WHERE WE LEFT OFF

# Last class in 60 seconds

### Five enemies, three moves

Missing · normalize · encode — and the type picks the encoding (no fake order!).

### Every drop is a claim

Cleaning decisions got reasons; disclosure became habit.

### Shortcut learning

The X-ray token: suspiciously good = a hypothesis to attack.

**Homework share:** the strangest thing you found while cleaning — and your shortcut audit. Your data is now clean and documented… **but it still can't speak.** Today we give it a voice: one conclusion, three honest charts.

AGENDA · HOW THIS HOUR RUNS

# Where we're going

- **0–8'** Homework share + four datasets in one disguise
- **8–18'** Charts are arguments · which chart for which job
- **18–28'** Correlation — and its favorite trap
- **28–38'** Honest visuals: the axis that lies · matplotlib's five lines
- **38–46'** Demos: charts in Colab · the AI chart critic
- **46–54'** Two labs: ASCII charts · Anscombe, verified by hand
- **54–60'** Your 3-chart data story, homework & wrap-up
- Quizzes & 💬 comments throughout

**By the end:** pick the right chart for a claim, compute and distrust correlation appropriately, spot an axis that lies, and tell one clear story with exactly three honest charts.

WARM-UP · ACTIVATE

# Quick check · tap an answer

**When is a self-driving car's vision least trustworthy — near or far? big objects or small? Where does that answer live?**

- In the car's user manual
- In expert opinion — ask three engineers
- Already in your detection log, waiting for the right chart to reveal it
- Nowhere; trustworthiness can't be measured

THE STORY

# Four datasets, one disguise

In 1973, statistician **Francis Anscombe** built four little datasets with nearly identical statistics — same mean, same variance, same correlation, same best-fit line. By the numbers they were **twins**. Then he plotted them. One was a clean line, one a curve, one had a single wild outlier, one was almost vertical. **The numbers hid everything; only the charts told the truth.**

🔊 **Speak this story** Neural voice · click to play / pause

You'll meet the actual four on the visual slide — and in Lab 2 you'll **verify his trick yourself**: compute the identical stats, then plot the four shapes. Fifty years later it's still the best two-minute argument for "always plot your data."

CONCEPT 1 · WHAT A CHART IS FOR

# A chart is an argument

### 📊 Histogram

"How is one variable **distributed**?" — where do my detection distances pile up?

### ✨ Scatter

"How do two variables **relate**?" — does confidence fall as objects shrink?

### 🔗 Correlation

"How **strong** is that relationship?" — one number (r) summarizing the scatter.

**KEY IDEA** — every chart should answer a question you can state **in one sentence**. If you can't say the sentence, you don't need the chart. (Recognize the discipline? It's Class 2's picture test — now pointed at output instead of input.)

CONCEPT 2 · THE QUESTION PICKS THE CHART

# Five jobs, five charts

### 📈 Trend over time

→ **line chart**. Sales by month, temperature by hour.

### ⚖️ Compare categories

→ **bar chart**. Detections per object class.

### 🥧 Parts of a whole

→ **pie** (sparingly!). Market shares — only when parts sum to 100%.

### 🏔️ Distribution

→ **histogram / box plot**. Ages, distances, scores.

### 🔀 Relationship

→ **scatter**. Confidence vs. size; study hours vs. grades.

### 🧭 The rule

State the question first — **the question picks the chart**. (Sound familiar? "The type picks the tools" — Class 7.)

CHECK · PICK THE CHART

# Quick check · tap an answer

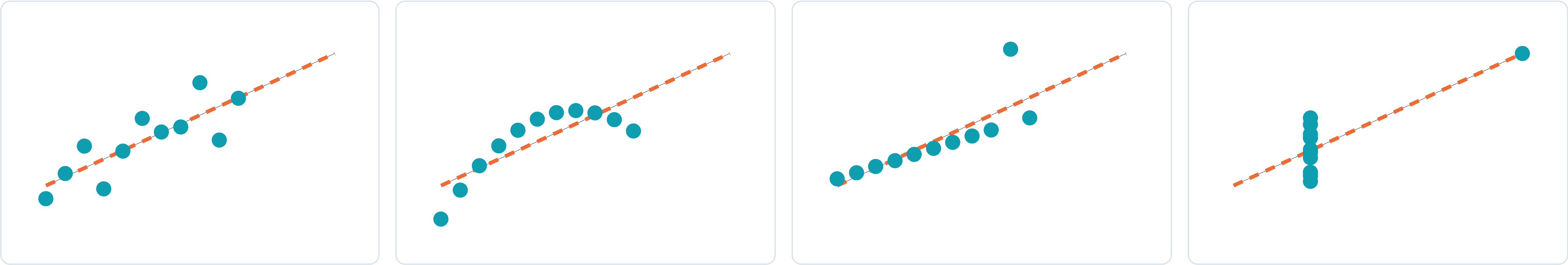
**Your claim: 'the detector's confidence drops as objects get smaller.' Which chart carries that argument?**

- A pie chart of object classes
- A scatter plot: box size on x, confidence on y
- A bar chart of average confidence
- A line chart of confidence over time

THE IDEA IN ONE PICTURE · THE REAL QUARTET

# Same stats, four different truths

These are Anscombe's actual numbers. All four: mean(x)=9, mean(y)=7.5, correlation r≈0.816, best-fit line y = 3 + 0.5x (dashed):



I · linear　　II · curved　　III · outlier　　IV · vertical

**Takeaway:** identical summary numbers can hide totally different shapes — a curve, a killer outlier, a vertical stack. **Always plot your data** before you trust any statistic about it. describe() is the beginning of looking, never the end.

CONCEPT 3 · THE STRENGTH NUMBER

# Correlation: one number for a relationship

### $r \to +1$

Rise together, tightly. Study hours vs. grades (usually!).

### $r \to 0$

No *linear* pattern. Shoe size vs. music taste.

### $r \to -1$

One rises, the other falls. Object size vs. distance in our log.

### ⚠️ Trap 1 · correlation ≠ causation

Ice-cream sales correlate with drownings — summer causes both. An r, however large, never proves one thing *drives* the other. (Class 5's Titanic quiz, formalized.)

### ⚠️ Trap 2 · r only sees straight lines

Anscombe's panel II: a perfect curve, r = 0.816 — same as the line! Strong nonlinear patterns can hide behind any r. The fix is always the same: **plot it**.

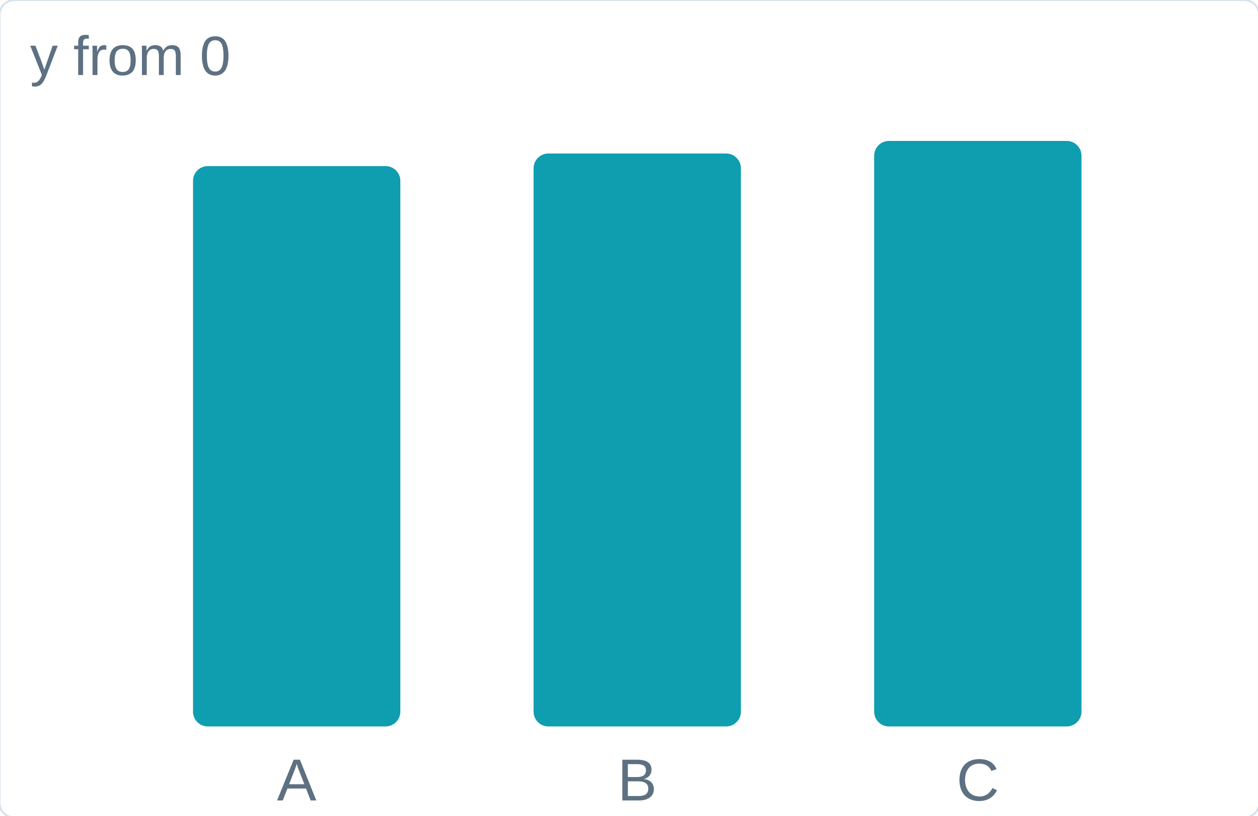
# Quick check · tap an answer

**Cities with more firefighters have more fire damage (strong positive r). Therefore firefighters cause damage. What went wrong?**

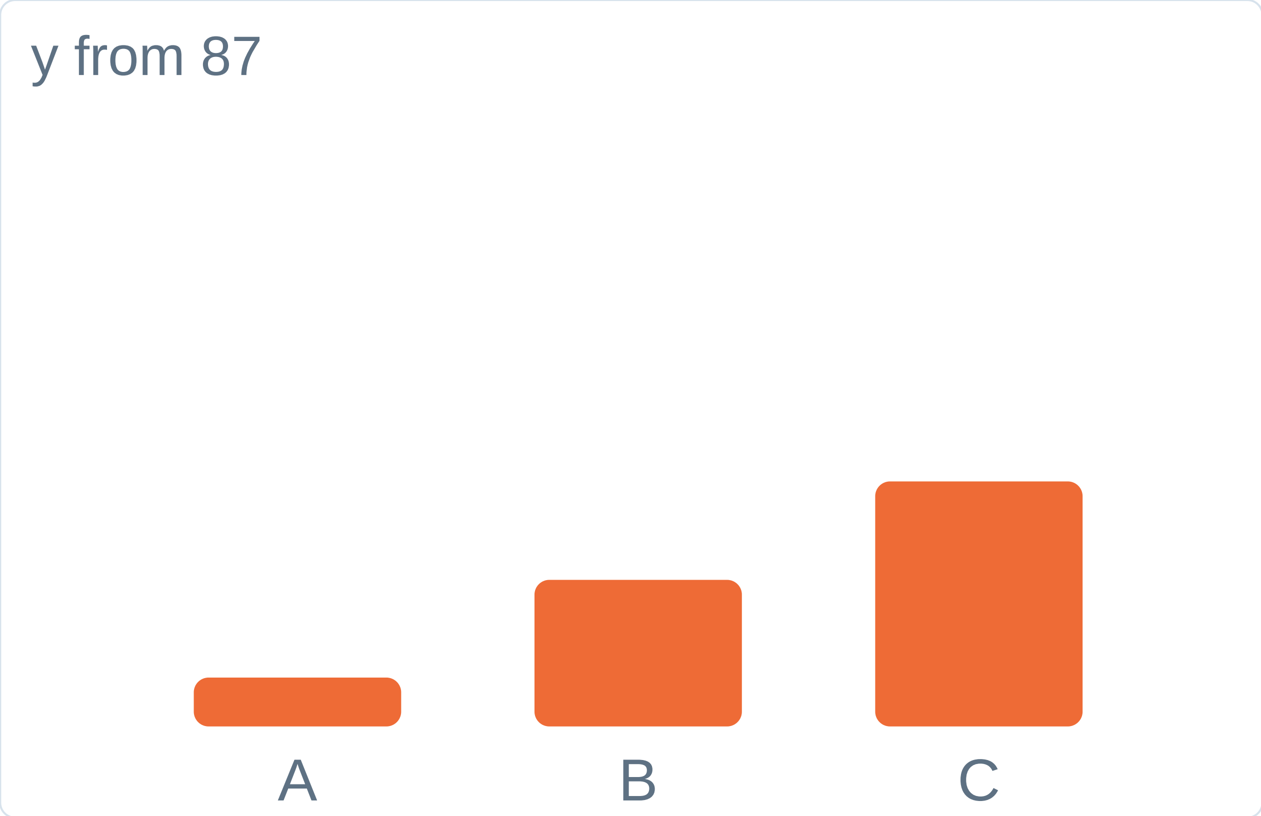
- Nothing — the data has spoken
- The r must have been computed incorrectly
- A lurking variable: bigger fires bring both more firefighters AND more damage — correlation isn't causation
- Firefighters should be reduced to test it

# The axis that lies

Same three detectors, same numbers — 88%, 90%, 92% accuracy. Watch the y-axis do the lying:




**Honest: axis starts at 0**
the three detectors are nearly tied




**Misleading: axis starts at 87**
C looks 3× better — same numbers!

**The honest-chart checklist:** axes start at sensible values (bars: usually 0) · units labeled · every point's meaning stated · **caption self-contained** — a reader should get the message without you narrating. Break these to make a point *look* bigger, and you've joined the X-ray token on the dark side.

THE TOOL · FIVE LINES TO A REAL CHART

# matplotlib, gently

## Histogram

```
import matplotlib.pyplot as plt
plt.hist(df["dist_m"], bins=20)
plt.title("Detection distances")
plt.xlabel("distance (m)")
plt.ylabel("count"); plt.show()
```

## Scatter

```
plt.scatter(df["box_area"],
            df["conf"], alpha=0.5)
plt.title("Confidence vs. size")
plt.xlabel("box area (px²)")
plt.ylabel("confidence"); plt.show()
```

**Notice what's NOT optional:** three of the five lines are title and axis labels — the same rule since your first Colab chart (Class 5). The plotting is one line; the **honesty** is the other four. Correlation, when you need the number: `df["a"].corr(df["b"])`.

CASE STUDY · OUR RUNNING EXAMPLE, ARGUED

# When is perception least confident?

1

### Histogram

Distances pile up at 5–20 m — but a long tail stretches past 60 m. *"Far detections exist and matter."*

2

### Scatter

Confidence vs. box size: a clear upward drift — small boxes, shaky confidence. *"Size drives certainty."*

3

### Correlation

r(size, conf) ≈ +0.7 · r(distance, conf) ≈ −0.6 — the numbers behind the pictures.

**Conclusion** ▸ Our detector is least trustworthy exactly where stakes are high: **small, far objects** — which is where a night-time pedestrian first appears.

**Anatomy of a data story:** each chart answers one sentence; together they corner a single conclusion — and that conclusion is Class 2's strong question, **now with evidence**. This is the story-shape your project copies today.

DISCUSS · 5 MINUTES · NUMBERS THAT HIDE

# A great summary number hiding a problem

Anscombe's mean hid four different shapes. What's a number from **your** world that could look great in a summary while hiding trouble underneath?

## Find one

A class average hiding a struggling half? An app's average rating hiding a review-bomb? A model's 95% accuracy hiding one always-missed class?

## Reveal it

Which chart exposes the hiding place? And — have *you* ever been fooled by a chart? What was the trick?

Pairs, 3 minutes — best hidden-number + revealing chart in 💬.

LIVE DEMO 1

# The data story, built live in Colab

From cleaned log to three captioned charts in ten minutes — the exact flow of tonight's deliverable.

```
You ▸ df["box_area"] = df.box_w * df.box_h # feature engineering, Class 8!
You ▸ plt.hist(df["dist_m"], bins=20) + title + labels → chart 1, captioned
You ▸ plt.scatter(df["box_area"], df["conf"], alpha=0.5) → chart 2, captioned
You ▸ df[["box_area","dist_m","conf"]].corr().round(2) → the numbers for chart 3
You ▸ plt.savefig("story1.png", dpi=150) — export for your poster/README
```

colab.research.google.com

LIVE DEMO 2

# The AI chart critic

Class 4's mentor hat, aimed at your charts: generate faster, then get audited for honesty.

**You** ▸ Write matplotlib code for a histogram of dist_m with proper title, axis labels, and units.

**You** ▸ Here's my chart code + caption: [paste]. Critique it against this checklist: honest axes, labeled units, self-contained caption.

**You** ▸ My claim is 'small objects get low confidence.' Which chart best supports it — and what chart would OVERSTATE it?

**You** ▸ Act as a skeptical reviewer: how could this chart mislead someone? Fix it.

chatgpt.com

# Build a histogram and a scatter — out of text

No library, no magic: a histogram is **counting into buckets**; a scatter is **placing points on a grid**. Build both from raw characters and the concepts become yours forever.

```
# The detection log's distances (m) and confidences - cleaned, Class 8 style
dist = [8.2, 15.0, 14.1, 22.4, 9.5, 13.0, 7.8, 16.2, 21.0, 6.9,
        31.5, 44.0, 12.3, 55.8, 18.7, 26.4, 38.2, 9.9, 48.5, 11.2]
conf = [0.94, 0.88, 0.91, 0.79, 0.62, 0.95, 0.58, 0.90, 0.83, 0.97,
        0.71, 0.55, 0.92, 0.42, 0.85, 0.74, 0.61, 0.96, 0.47, 0.93]

# ── HISTOGRAM = count into buckets ──────────────────────────────
print("Distance distribution (each # = 1 detection)")
for lo in range(0, 60, 10):
    n = sum(lo <= d < lo + 10 for d in dist)
    print(f"{lo:2d}-{lo+10:<3d}m | {'#' * n}")
```

▶ Run · 🎧 Explain each line · ✨ Explain · 🐞 Debug · ⚡ Improve · 🤖 Mira

Python loads on first Run

```
▶ Run the code to see the output here.
```

HANDS-ON LAB 2 · CHECK A LEGEND'S HOMEWORK

# Anscombe's quartet, verified by hand

Don't take the story on faith. Compute the stats yourself — means, correlation, fit line — then plot the four shapes. **The numbers say twins; your eyes will disagree.**

```
# Anscombe's actual 1973 numbers - all four sets
x123 = [10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5]
sets = {
 "I  ": (x123, [8.04,6.95,7.58,8.81,8.33,9.96,7.24,4.26,10.84,4.82,5.68]),
 "II ": (x123, [9.14,8.14,8.74,8.77,9.26,8.10,6.13,3.10,9.13,7.26,4.74]),
 "III": (x123, [7.46,6.77,12.74,7.11,7.81,8.84,6.08,5.39,8.15,6.42,5.73]),
 "IV ": ([8,8,8,8,8,8,8,19,8,8,8],
         [6.58,5.76,7.71,8.84,8.47,7.04,5.25,12.50,5.56,7.91,6.89]),
}

def mean(v): return sum(v) / len(v)
def corr(xs, ys):                      # Pearson's r, from scratch
```

▶ Run | 🎧 Explain each line | ✨ Explain | 🐞 Debug | ⚡ Improve | 🤖 Mira

Python loads on first Run

```
▶ Run the code to see the output here.
```

YOUR PROJECT

# A 3-chart data story

Use **exactly three charts** to make **one clear, defensible conclusion** about your own dataset.

- State your conclusion in **one sentence** — first, before any plotting
- Pick the 3 charts that genuinely support it (the question picks the chart!)
- Build them with titles, labeled axes, units
- Write a **self-contained caption** for each

## Deliverable

A "data story" poster: 3 captioned charts + the one-sentence conclusion.

## Success looks like

The 3 charts genuinely support the one claim — and axes and captions are honest and clear.

BUILD IT NOW · HANDS-ON

# Conclusion first, charts second

1. Open your Class-8 Colab (cleaned data ready) → write your **one-sentence conclusion** as a markdown cell
2. Chart 1 — usually a **histogram**: show the landscape your claim lives in
3. Chart 2 — usually a **scatter**: show the relationship doing the work
4. Chart 3 — the **strength or comparison**: .corr(), or a grouped bar
5. Caption each in one sentence: *what it shows* → *why that supports the claim*
6. The honesty pass: axes, units, labels — then **Run all**, savefig, commit to GitHub

Colab | matplotlib | ChatGPT / Claude (chart critic) | GitHub

**The cut rule:** if a chart doesn't pull its weight for the claim, cut it — **three strong beats five weak**. (And a chart you can't caption in one sentence is telling you something.)

BENCHMARK · JUDGE AGAINST THIS

# What a finished data story looks like

**Conclusion:** *"Our detector is least reliable on small, distant objects — exactly where night-time pedestrians first appear."*

### 📊 Chart 1 · histogram

Distances, 20 bins, x-labeled in metres. **Caption:** "30% of detections are beyond 25 m — the far zone isn't rare, so failures there matter."

### ✨ Chart 2 · scatter

Box area vs. confidence, α=0.5, both axes united. **Caption:** "Confidence sinks with size — below ~2,000 px², most detections fall under 0.7."

### 🔗 Chart 3 · correlation

r table: size↔conf +0.72, distance↔conf −0.61. **Caption:** "The drift is strong, not a squint: size and distance together explain most confidence loss."

**The bar:** each caption is *what it shows* → *why it supports the claim*, the three charts hit distribution → relationship → strength, and nothing needs the author standing next to it. This poster is slide 3 of your final presentation (Class 15), finished seven weeks early.

REFERENCE · KEY TERMS

# Today's vocabulary

**Histogram** — counts in buckets: the shape of one variable's distribution. Bin width is a judgment call.

**Scatter plot** — two variables, one point per row: the shape of a relationship.

**Correlation (r)** — linear-relationship strength, −1…+1. Blind to curves; silent on causes.

**Lurking variable** — the hidden driver behind a correlation (summer → ice cream + drownings).

**Truncated axis** — a y-axis that skips zero to inflate differences: the most common chart lie.

**Data story** — one conclusion, a few honest charts, self-contained captions. Your deliverable — and Class 15's opening act.

RECOMMENDED HOMEWORK

# Before next class

1. **Finish your 3-chart data story** (conclusion + captioned charts, honesty pass done) — submit the Colab link in Google Classroom, commit the PNGs to GitHub.
2. **Chart-lie hunt:** find one misleading chart in the wild (news, ads, social media). One sentence: what's the trick, and what's the one-line fix?
3. **Re-run Lab 2's challenge:** remove set III's outlier and report what happens to r — one sentence on what that teaches about outliers and correlation.

## ⭐ Stretch (optional)

Meet the **Datasaurus Dozen** — Anscombe's modern heir: thirteen datasets with identical stats, one of which is a *dinosaur*. Scroll the animation and grin.

autodesk research · Datasaurus

The chart-lie hunt is the habit that outlives the course — once you see truncated axes, you can't unsee them. Questions → 💬 on any slide.

TODAY'S LINE TO REMEMBER

# "The greatest value of a picture is when it forces us to notice what we never expected to see."

— John Tukey, father of exploratory data analysis

Anscombe proved it with four plots. Tonight, your three charts might surprise **you** first.

🔊 Speak this

WRAP-UP

# Three things to keep

- Charts make **arguments** — every chart answers a one-sentence question
- **Always plot your data** — identical stats can hide four different truths
- Honest axes, labeled units, self-contained captions — **or it's decoration**

**Exit ticket:** the one-sentence conclusion of your data story — post it in 💬.

Next · Class 10 — Stage 4 begins: your first model that **predicts**. The data stage is complete.

BACKUP MATERIAL

# Extra slides — if there's time

Deeper dives and extra practice. Skip in a tight hour; pull up on demand or for a longer session.

**The liar's gallery**

three classic chart tricks

**The extended chart menu**

box plots, lines, heatmaps

**Bonus quiz**

one more check

BACKUP · THE LIAR'S GALLERY

# Three classic chart tricks (and the catch)

### ✂️ The truncated axis

y starts at 87, differences triple visually. **Catch:** read the axis numbers *first*, always.

### 🪟 The cherry-picked window

"Sales up 40%!" — chart starts at last year's crash. **Catch:** ask what happens if the x-axis extends left.

### ⚖️ The double y-axis

Two unrelated scales overlaid until they "correlate." **Catch:** two y-axes = two stories; demand them separately.

**The general antidote:** a chart is a claim (concept 1) — so apply Class 4's discipline to it: *who made this, what's the source, what would the honest version look like?* Verification isn't just for text.

BACKUP · THE EXTENDED MENU

# Three more charts worth knowing

### 📦 Box plot

describe() as a picture: median, quartiles, and outliers flagged automatically. Best for comparing distributions across groups — `df.boxplot(column="conf", by="object")`.

### 📈 Line chart

The trend specialist — but only when x is *ordered* (time, distance). Lines on unordered categories draw connections that don't exist.

### 🔥 Heatmap

A whole correlation table at a glance — `plt.imshow(df.corr())`. The instant map of which variables move together.

Still the same rule: **the question picks the chart.** New chart types are new answers, not new decorations — add one to your story only if it carries a sentence the core three can't.

BACKUP · BONUS CHECK

# Quick check · tap an answer

**A news chart shows crime 'exploding' — a steep red line. You check: the y-axis runs 41.0 to 42.5 incidents, and the x-axis covers exactly the two worst months of a flat year. Diagnose the tricks.**

- No tricks — steep lines mean big changes
- One trick: bad colors
- Two stacked tricks: truncated axis magnifying a ~3% wiggle, plus a cherry-picked time window
- The data must be fake